Venn Diagrams and Contingency Tables

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Events and sample spaces.
- **Leads to:** Joint and marginal probability.

1. Overview

Key point: A **Venn diagram** (G-2086) draws events as overlapping circles inside the sample space; a contingency table holds the same counts in a grid. Both show at a glance which outcomes two events share.

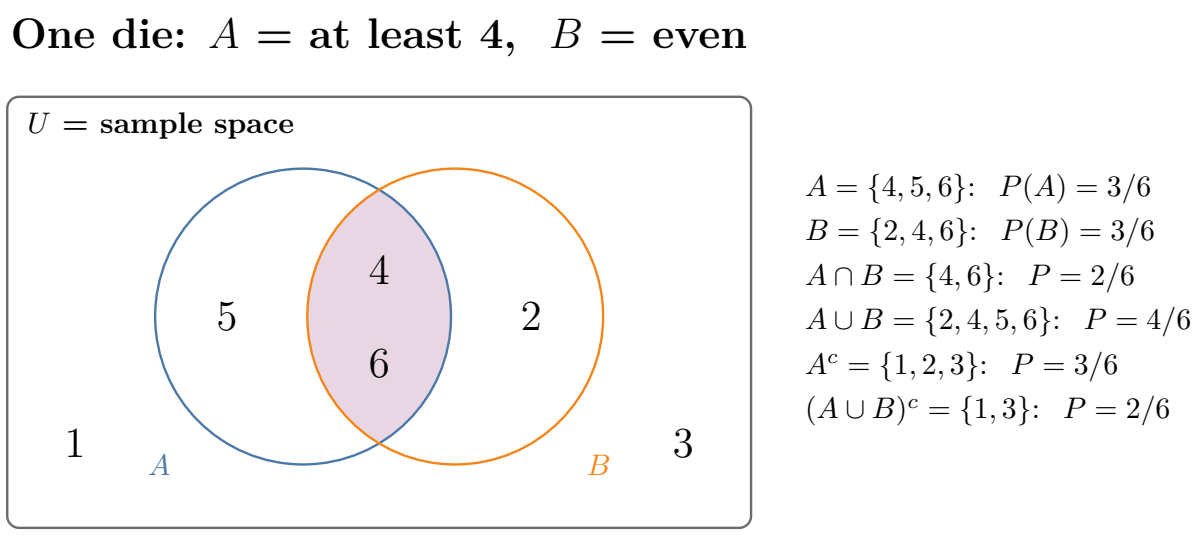


Figure 1: One die with two events, A = at least 4 and B = even, as a Venn diagram

The **sample space** is the set of all possible outcomes of an experiment; for one die it is the six numbers 1 to 6. An **event** is a set of outcomes, such as "even" $= \{2, 4, 6\}$. $P(A)$ is the probability of event A. Figure 1 rolls one fair die and draws two events: $A =$ "at least 4" and $B =$ "even". Every outcome sits in exactly one region: in A only, in both, in B only, or in neither. Counting the outcomes in a region gives its probability: with six equally likely outcomes, a region holding 3 of them has probability 3/6.

This Note covers two ways of showing how events overlap:

- the Venn diagram, a picture (section 2);
- the **contingency table** (G-464), the same information as a grid of counts (section 3);
- how to switch between them (section 4).

Joint, marginal and conditional probabilities, the subject of the joint, marginal and conditional probability, are all read off these two views.

Sample spaces and events are defined in the random experiments and events; the **complement** (G-422) and addition rules used here are in the empirical and theoretical probability.

2. Venn diagrams

> **Key point:** In a Venn diagram the rectangle is the whole sample space, with probability 1; each event is a circle, and overlaps show shared outcomes.

A **Venn diagram** (G-2086) comes from set theory. A rectangle stands for the **universal set** (G-2053) U, everything there is. Each event is drawn as a circle inside it.

2.1 The rectangle is the sample space

> **Key point:** In probability, the **universal set** (G-2053) is the sample space, so $P(U) = 1$.

In probability the universal set is the sample space of the experiment. For one die, the rectangle holds the outcomes 1 to 6.

Every trial gives some outcome inside the rectangle, so the rectangle is the sure event:

$$P(U) = 1$$

2.2 One event and its complement

> **Key point:** A circle marks one event; everything in the rectangle outside the circle is its **complement** (G-422).

Take the event $A =$ "a number of at least 4" $= \{4, 5, 6\}$ (the braces list the outcomes that belong to the event). Its outcomes go inside a circle; 1, 2 and 3 stay outside it, in the rectangle.

- $P(A)$ counts three of the six equally likely outcomes:

$$P(A) = 3/6 = 1/2$$

- The region outside the circle is the complement $A^c = \{1, 2, 3\}$. Its probability is the whole rectangle minus the circle (the complement rule):

$$P(A^c) = P(U) - P(A) = 1 - \frac{1}{2} = \frac{1}{2}$$

2.3 Two events and their four regions

> **Key point:** Two overlapping circles split the sample space into four regions: A only, both, B only and neither; each union, intersection or complement is a set of these regions.

Add a second event, B = "an even number" $= \{2, 4, 6\}$. The outcomes 4 and 6 belong to both events, so they go in the overlap of the two circles (Figure 1):

Region	Outcomes	Probability
A only	5	1/6
both A and B	4, 6	2/6
B only	2	1/6
neither	1, 3	2/6

The four probabilities add up to 1, because every outcome lies in exactly one region:

$$\frac{1}{6} + \frac{2}{6} + \frac{1}{6} + \frac{2}{6} = \frac{6}{6}$$

$$\frac{6}{6} = 1$$

Figure 2 shades the four events built from these regions.

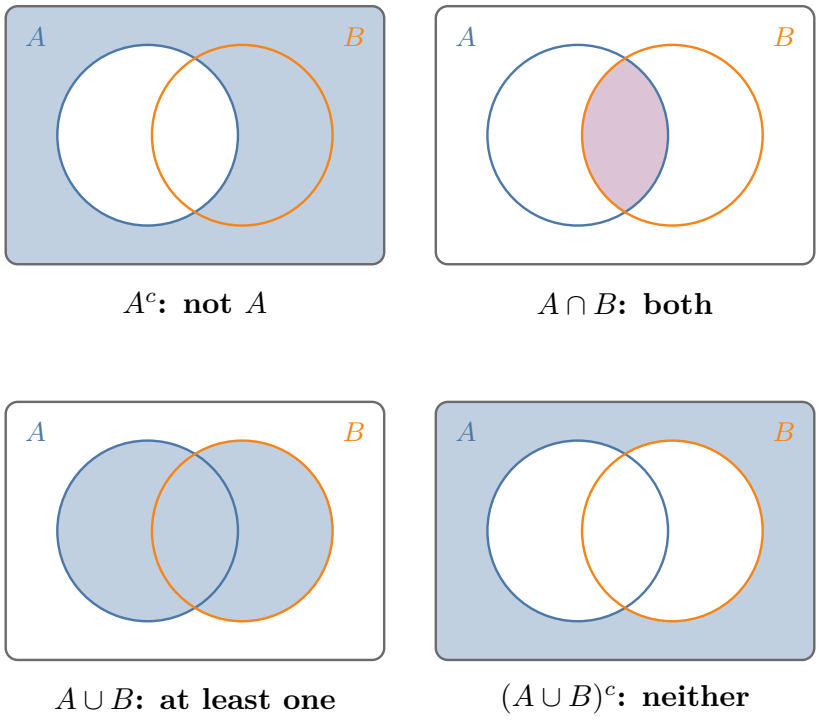


Figure 2: Four events as shaded regions of a two-circle Venn diagram

Each one is read from the diagram by counting the outcomes in its shaded part:

- **Intersection** (G-967) $A \cap B$, the overlap: $\{4, 6\}$.

$$P(A \cap B) = 2/6$$

- **Union** (G-2045) $A \cup B$, everything inside either circle: $\{2, 4, 5, 6\}$.

$$P(A \cup B) = 4/6$$

- **Complement** (G-422) A^c, everything outside circle A: $\{1, 2, 3\}$.

$$P(A^c) = 3/6$$

- **Neither**, $(A \cup B)^c$, everything outside both circles: $\{1, 3\}$.

$$P(\text{neither}) = 2/6$$

The union can also be found without counting, by the general addition rule from the empirical and theoretical probability. The overlap is inside both circles, so it is subtracted once:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = \frac{3}{6} + \frac{3}{6} - \frac{2}{6}$$

$$P(A \cup B) = \frac{4}{6}$$

"Neither" is the complement of the union, so it follows from the union:

$$P\Big((A \cup B)^c\Big) = 1 - P(A \cup B)$$

$$P\Big((A \cup B)^c\Big) = 1 - \frac{4}{6}$$

$$P\Big((A \cup B)^c\Big) = \frac{2}{6}$$

Both routes agree with the counts in the table above.

Extra: "Neither A nor B" can also be built as "not A and not B":

$$A^c = \{1, 2, 3\}$$
$$B^c = \{1, 3, 5\}$$
$$A^c \cap B^c = \{1, 3\}$$

The identity is **De Morgan's law** (G-550): $(A \cup B)^c = A^c \cap B^c$. Its twin, $(A \cap B)^c = A^c \cup B^c$, says "not both" means "at least one of them fails". In Figure 2, the panel labelled "neither" is exactly this region outside both circles.

Extra: A **Venn diagram** (G-2086) shows overlap, not independence. Mutually exclusive events (see the **mutually exclusive events** (G-1286)) are circles that do not touch. Whether two overlapping events are independent needs a calculation. If they were independent, the overlap would equal the product:

$$P(A) \cdot P(B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

But the overlap is:

$$P(A \cap B) = \frac{2}{6} = \frac{1}{3}$$

The two differ, so A and B are dependent (see the **independent events** (G-934)): knowing the roll is even makes "at least 4" more likely: of the three even outcomes 2, 4 and 6, two are at least 4, so the chance rises from 1/2 to 2/3.

3. Contingency tables

> **Key point:** A contingency table puts the categories of one event in the rows and of the other in the columns; each inner cell counts the outcomes in one Venn region, and the totals count whole circles.

A **contingency table** (G-464), also called a **crosstab** (G-511; see the bivariate and multivariate analysis), counts every pair of categories of two categorical **features** (G-772; a feature is a variable, one column of the data table). Applied to two events, it holds the same information as the **Venn diagram** (G-2086), arranged as a grid.

3.1 The die as a table

> **Key point:** Two yes/no events give a 2 by 2 table; its four inner cells are the four Venn regions.

For the die, the rows ask "is the number at least 4?" and the columns ask "is it even?". Each cell counts the outcomes that answer both questions that way:

	even	odd	total
at least 4	2 (4, 6)	1 (5)	3
below 4	1 (2)	2 (1, 3)	3
total	3	3	6

Each inner cell matches one region of Figure 1: "at least 4 and even" is the overlap, "below 4 and odd" is the outside. The row totals count circle A and its **complement** (G-422); the column totals count circle B and its complement.

Each yes/no question works like a random variable with two values, 1 for yes and 0 for no (see the random variables as functions). The table lists every combination of the two.

3.2 A class of 100 students

> **Key point:** When counts come from data, the table is filled from the facts given; the cell nobody mentions is what is left of the total.

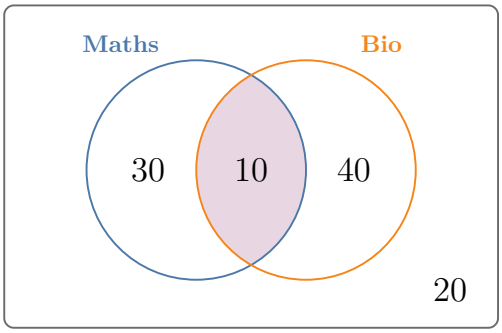
A class has 100 students. 40 chose only Bio as their major, 30 chose only Maths, and 10 chose both. How many chose neither?

The table makes the missing count easy to find. The "neither" cell is whatever remains of the 100:

$$100 - 30 - 40 - 10 = 20$$

Figure 3 shows the counts both ways.

100 students: the same counts, two views



	Bio	not Bio	total
Maths	10	30	40
not Maths	40	20	60
total	50	50	**100**

Figure 3: The 100 students as a Venn diagram (left) and a contingency table with totals (right)

Reading the table:

- **Inner cells:** 10 took both, 30 Maths only, 40 Bio only, 20 neither.
- **Row totals:** 40 took Maths (10 + 30), 60 did not.
- **Column totals:** 50 took Bio (10 + 40), 50 did not.
- **Grand total:** 100, the whole class.

The row and column totals are written in the margins of the table. They return as the counts behind marginal probabilities.

3.3 From counts to probabilities

> **Key point:** Dividing every cell by the grand total turns the table into probabilities for one randomly chosen student.

Pick one student at random. Dividing each count by 100 gives the probability of each cell and each total:

	Bio	not Bio	total
Maths	0.10	0.30	0.40
not Maths	0.40	0.20	0.60
total	0.50	0.50	1.00

Now every probability from section 2 can be read from the table:

- $P(\text{Maths} \cap \text{Bio}) = 0.10$, one inner cell.
- $P(\text{Maths}) = 0.40$, a row total.
- The **union** (G-2045) uses the general addition rule:

$$P(\text{Maths} \cup \text{Bio}) = 0.40 + 0.50 - 0.10$$

$$P(\text{Maths} \cup \text{Bio}) = 0.80$$

- The **complement** (G-422) of the union:

$$P(\text{neither}) = 1 - 0.80 = 0.20$$

Figure 4 shades the cells behind each of these four probabilities: one cell for the **intersection** (G-967), a row for $P(\text{Maths})$, three cells for the union, and the one remaining cell for "neither".

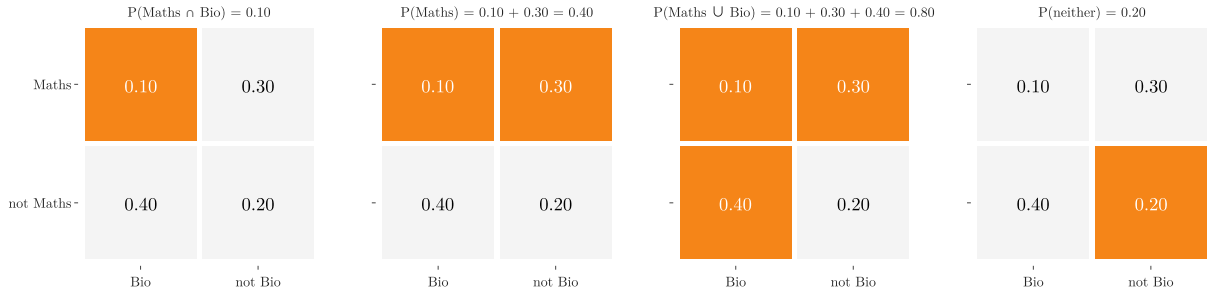


Figure 4: The students' probability table four times; the orange cells add up to the probability named above each panel.

4. Switching between the two views

> **Key point:** Venn regions and table cells are the same sets of outcomes, so either view can be rebuilt from the other.

The **Venn diagram** (G-2086) and the contingency table carry exactly the same information. Figure 5 lights up each Venn region of the die example together with its table cell, then a whole circle together with its total:

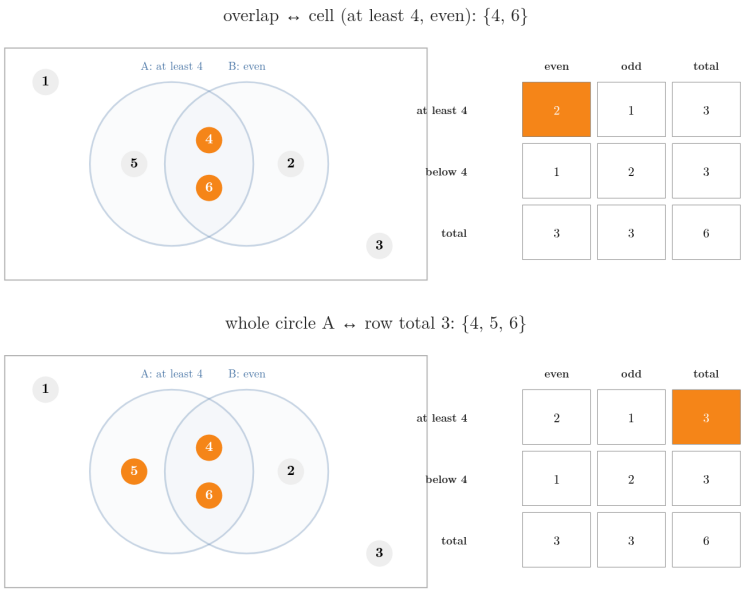


Figure 5: The die example: each Venn region (orange numbers) lights up with the matching cell of the contingency table; a whole circle matches a row or column total.

The full correspondence:

Venn diagram	Contingency table
rectangle U	grand total
overlap of the circles	cell (yes, yes)
part of A outside B	cell (A yes, B no)
part of B outside A	cell (A no, B yes)
outside both circles	cell (no, no)
whole circle A	row total for A yes
whole circle B	column total for B yes

Each view suits different work:

- A **Venn diagram** shows how events overlap at a glance, and makes **unions** (G-2045), complements and De Morgan's law easy to see.
- A **contingency table** scales to real data with many **observations** (G-1374; records, one row of the data table each) and more than two categories, such as three passenger classes. pandas builds it in one call, which the joint, marginal and conditional probability uses on the Titanic data.

5. Summary

Event	Venn region	Die (A: at least 4, B: even)	Students (Maths, Bio)
$A \cap B$	overlap	$\{4, 6\}$, 2/6	0.10
$A \cup B$	inside either circle	$\{2, 4, 5, 6\}$, 4/6	0.80
A^c	outside circle A	$\{1, 2, 3\}$, 3/6	0.60
$(A \cup B)^c$	outside both	$\{1, 3\}$, 2/6	0.20
U	whole rectangle	$\{1, \ldots, 6\}$, 1	1

- Each event in the table is a set of regions, so its probability is read by counting the outcomes in its shaded part.
- The rectangle is the sample space; its probability is 1, because every trial gives some outcome inside it.
- Two events split the sample space into four regions, whose probabilities add to 1, because every outcome lies in exactly one region. So a count nobody mentions ("neither") is what is left of the total.
- A **contingency table** (G-464) holds the same four regions as cells, and the circles as row and column totals, so either view can be rebuilt from the other; the table also scales to real data with many observations and categories.
- Dividing a table of counts by its grand total turns it into a table of probabilities, the probabilities for one randomly chosen member.
- Both views show at a glance which outcomes two events share, and joint, marginal and conditional probabilities are all read off them.

6. Sources

Built from

- CampusX, "Mastering Probability for ML: Joint | Marginal | Conditional Probability and Bayes' Theorem", YouTube, https://www.youtube.com/watch?v=ndHDsvqmbuI

7. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Venn diagram (G-2086)	A picture of events as circles inside a rectangle; overlaps show shared outcomes.
Contingency table (G-464)	A table that counts the observations for every pair of categories of two categorical features, so we can see whether the two are related; also called a crosstab.

Term	Meaning
Universal set (U) (G-2053)	The rectangle of a Venn diagram; in probability, the sample space, with $P(U) = 1$.
De Morgan's law (G-550)	Two rules for "not" over "or" and "and": "neither A nor B" is "not A and not B", $(A \cup B)^c = A^c \cap B^c$; "not both" is "at least one fails", $(A \cap B)^c = A^c \cup B^c$.
Grand total (G-867)	The sum of every cell of a contingency table: the size of the whole sample.
Row and column totals (G-1711)	The sums in the margins of a contingency table; each counts one whole event.
Complement (A^c) (G-422)	The event that A does not happen: every outcome not in A.
Intersection (A ∩ B) (G-967)	The set of elements that are in both A and B, such as the points inside both of two half-planes; for events, the event that both A and B happen.
Union (A ∪ B) (G-2045)	The set of elements in A or B (or both), such as the points inside either of two convex pieces of an obstacle; for events, the event that A or B (or both) happens.
Mutually exclusive events (G-1286)	Events that cannot happen at the same time; their intersection has probability 0.
Independent events (G-934)	Events where one happening does not change the probability of the other; for them the probability that both happen is the product of their probabilities.
Crosstab (G-511)	A table that counts the rows for every pair of categories of two categorical columns, so we can see whether the two are related; pandas builds it with `pd.crosstab`. Also called a contingency table.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Observation (G-1374)	One record of a dataset: one row of the data table.